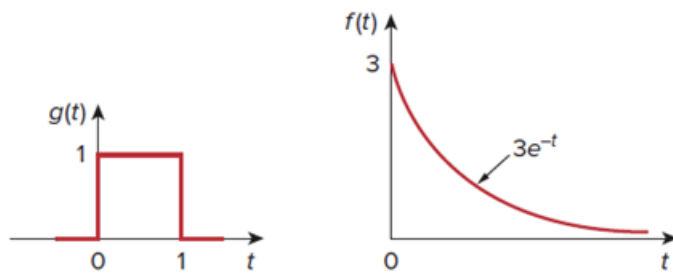


Problem

Given $g(t)$ and $f(t)$ in Fig. 15.20, graphically find $y(t) = g(t) * f(t)$.

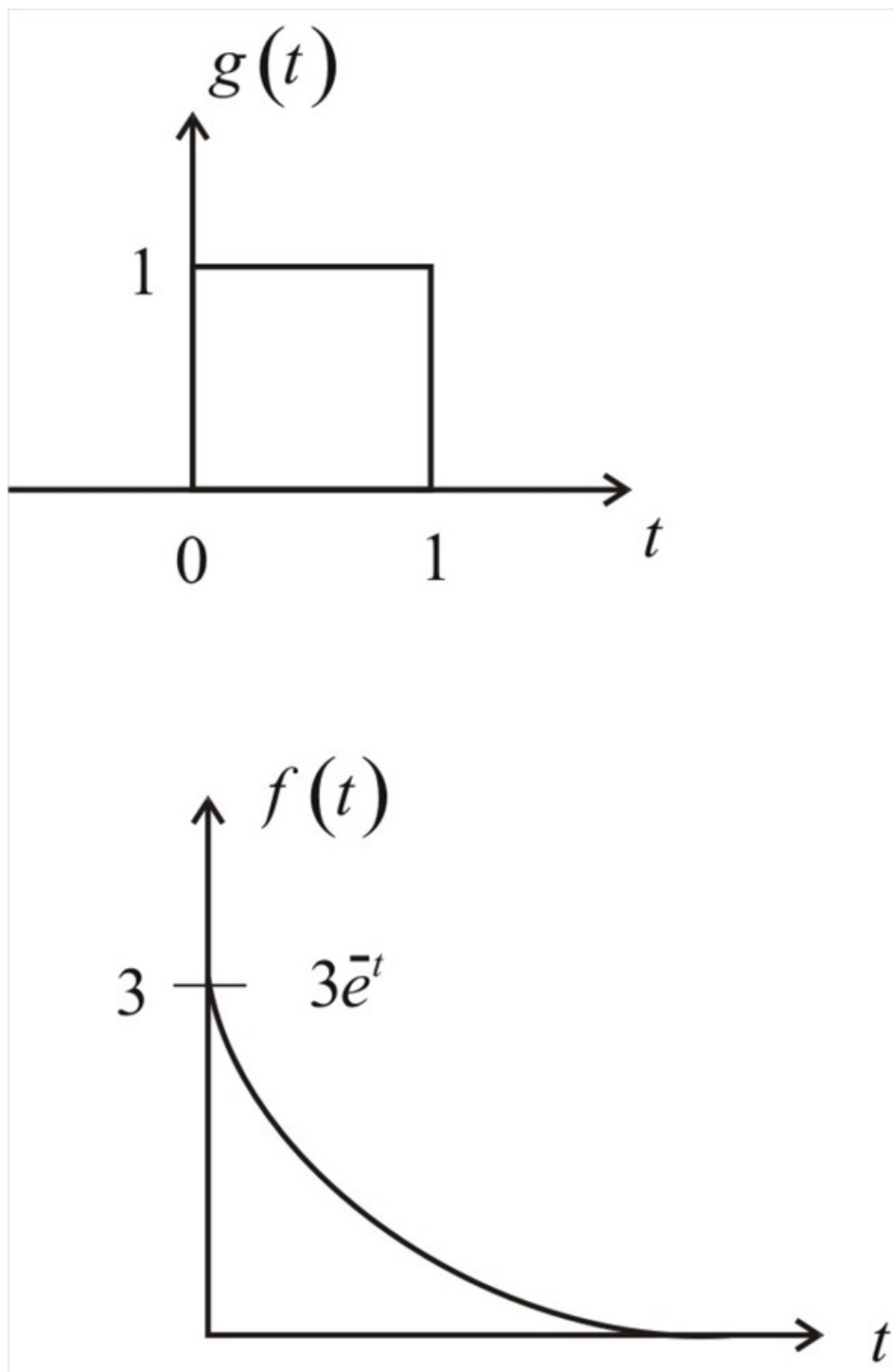
Figure 15.20:



Step-by-step solution

Step 1 of 6

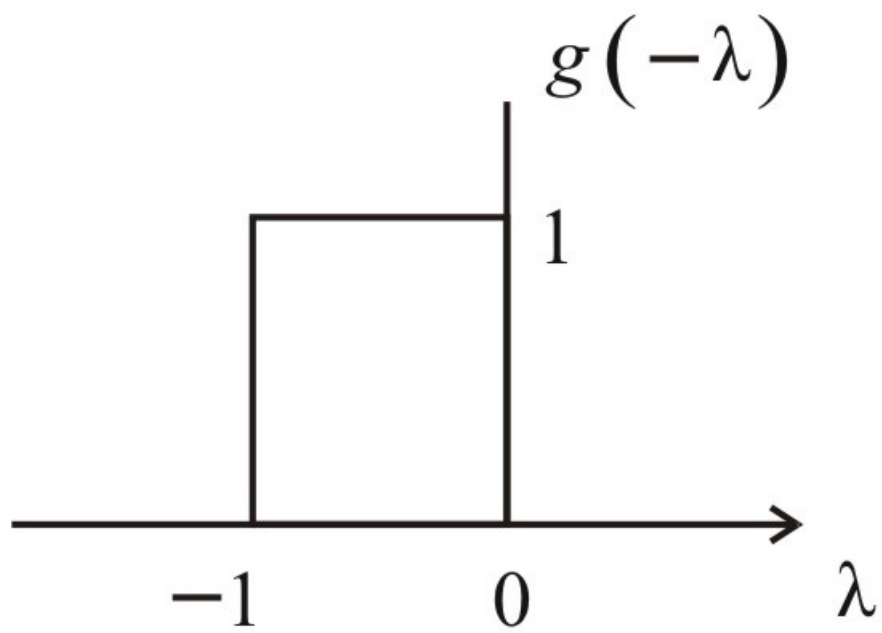
Given graphical functions are



Step 2 of 6

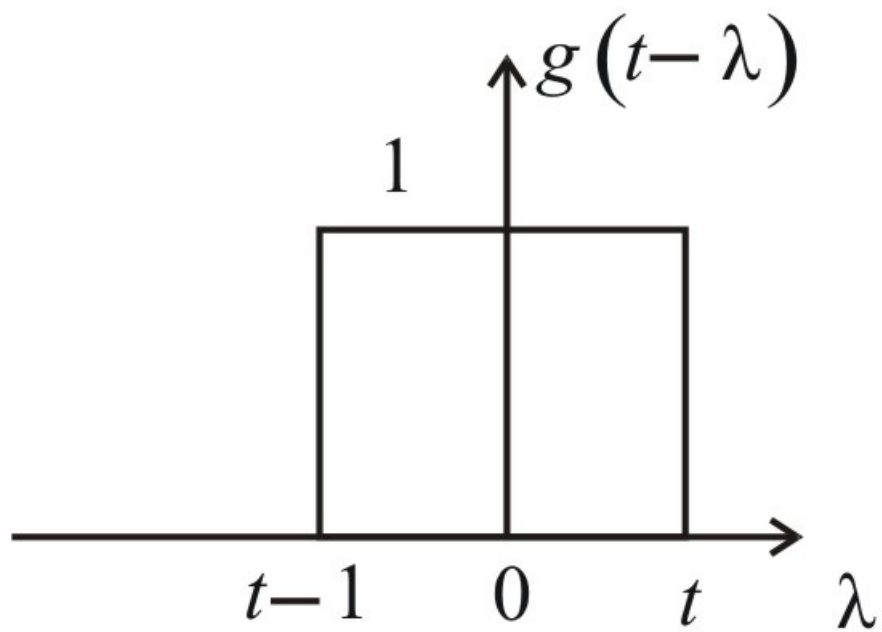
To find the $y(t) = g(t) * f(t)$

We fold $g(t)$ as



Step 3 of 6

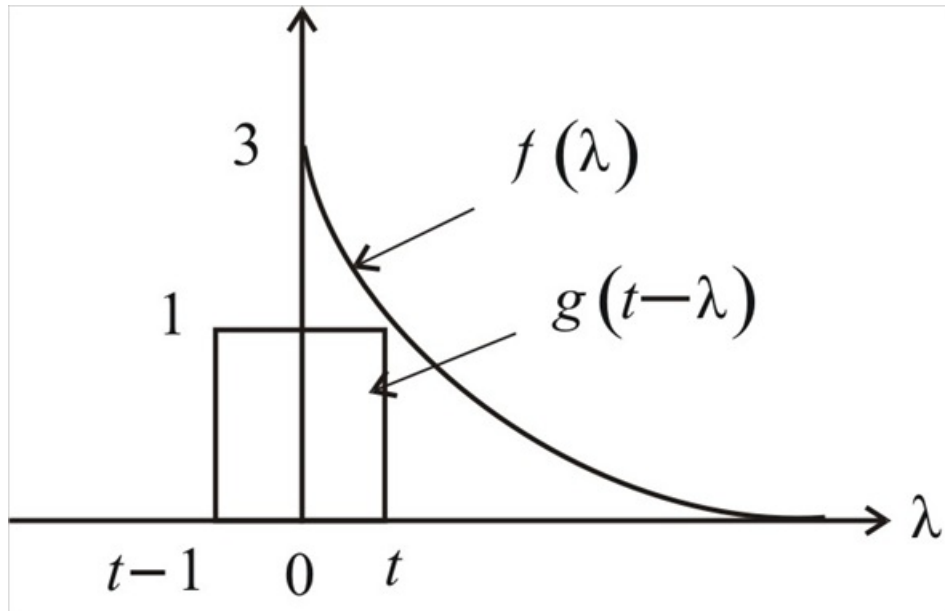
Then shift it by t ,



Step 4 of 6

There is no overlap of the two functions when $t < 0$ so that $y(0) = 0$

For $0 < t < 1$, $g(t-\lambda)$ and $f(\lambda)$ overlap from 0 to t ,



Step 5 of 6

Therefore,

$$y(t) = \int_0^t (1)(3e^{-\lambda}) d\lambda$$

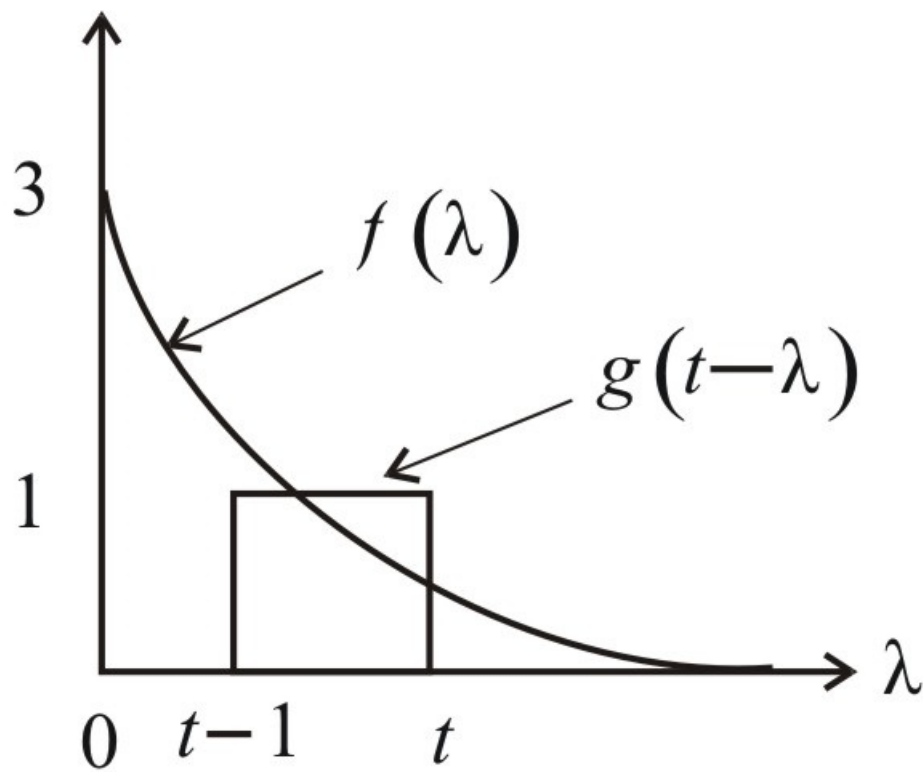
$$y(t) = \left. \frac{3e^{-\lambda}}{-1} \right|_0^t$$

$$y(t) = -3(e^{-t} - 1)$$

$$y(t) = 3(1 - e^{-t}) \quad 0 \leq t \leq 1$$

Step 6 of 6

For $t > 1$, the two functions overlap completely between $(t-1)$ and t , then



Therefore,

$$y(t) = \int_{t-1}^t (1)(3e^{-\lambda}) d\lambda$$

$$y(t) = \left. \frac{3e^{-\lambda}}{-1} \right|_{t-1}^t$$

$$y(t) = -3[e^{-t} - e^{-(t-1)}]$$

$$y(t) = 3e^{-t}[e-1] \quad t \geq 1$$

Therefore,

$$y(t) = \begin{cases} 3(1-e^{-t}), & 0 \leq t \leq 1 \\ 3e^{-t}[e-1] & t \geq 1 \\ 0 & \text{else where} \end{cases}$$